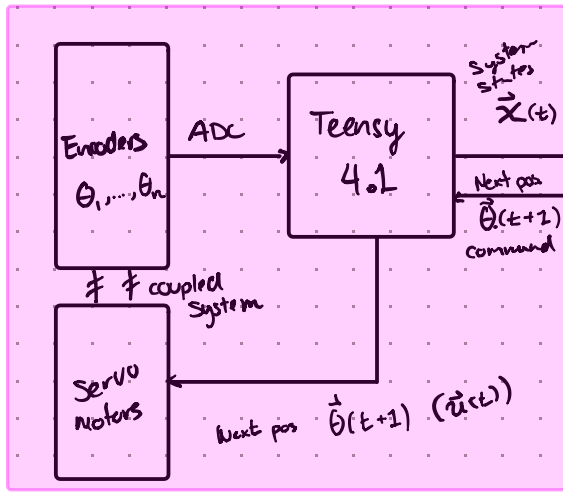
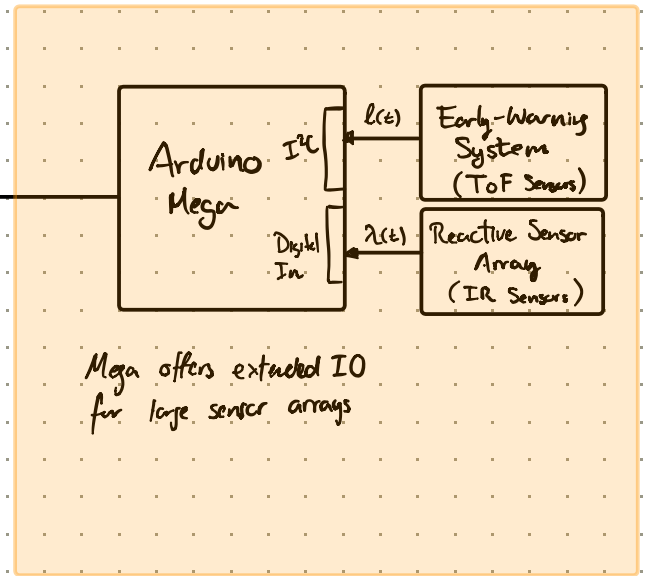
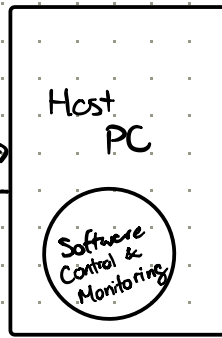


Hardware Loop:



Robotic System

Teensy offers more CPU power in comparison to other microcontroller units.



Sensor Fusion

Control System v1.0

Let:

$\hat{\theta}_m$: Measured states

$\hat{\theta}_r$: Reference states

$\hat{\lambda}$: IR Sensor states (Binary)

$\hat{l}$: ToF Sensor distances

And $\hat{\theta} = \{ \underbrace{\theta_1, \theta_2, \dots, \theta_n}_{\text{Servo pos}}, \underbrace{\phi, \omega}_{\text{Stepper States}} \}$

